When Can a Machine Judge Like a Human? Evaluating LLM-as-a-Judge Approaches for Human-Centered Computing Research

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ABSTRACT

Large language models (LLMs) are flooding into Human-Centered Computing (HCC) research as substitutes for human judgment: coding qualitative data, annotating crowdsourced corpora, simulating study participants, generating personas, and evaluating designs. Yet HCC's methods carry epistemological commitments (interpretation, reflexivity, positionality) that make uncritical substitution risky, and the field lacks an evidence-based account of *when* an "LLM-as-a-judge" can stand in for a human and when it cannot. We address this with four studies. **Study 1a** is a PRISMA systematic review of 217 studies across 29 domains that establishes the general reliability landscape of LLM-as-a-judge: substitution is validated (≈ or above the human-human ceiling) for objective, verifiable, low-expertise tasks judged in aggregate, and unreliable for subjective, expert, low-resource, or adversarial ones (mean reliability 3.15/5; only 31% of studies in the reliable band). **Study 1b** asks whether this landscape is a moving target: we re-run a flagship "promising" benchmark (SummEval) with the most powerful open models served via Clemson's vLLM API and ask whether they now *surpass* the agreement numbers reported for proprietary judges, distinguishing a "matter of time" from a structural ceiling. **Study 2** is a PRISMA systematic review of 55 studies focused on HCC, HCI, and qualitative-research methods that historically require human input. It finds that the field overwhelmingly positions LLMs to *augment* rather than replace humans (36 augment vs. 14 replace vs. 5 simulate), with an epistemic stance dominated by *tension* (38/55); objective crowd-annotation substitution is validated, interpretive qualitative analysis is reliable only as human-in-the-loop augmentation, and *simulating* participants, personas, and usability evaluations is the least reliable (flattening of identity groups, caricature, hallucinated usability issues). **Study 3** closes the gaps the first two studies expose, by running a standardized LLM-as-a-judge meta-evaluation, using open models served via Clemson's vLLM API and the agreement-against-human-ceiling methodology distilled from Studies 1-2, on HCC datasets that contain human labels but have never been formally tested with an LLM judge. Across 15000 judgments (a 6-model open panel, 9B-754B), it finds the human-agreement ceiling governs reliability: the panel matches or exceeds humans on low-ceiling NLI but falls short of, and fails the alt-test for, subjective social-computing annotation (emotion, hate speech), with binary offensiveness the strongest case. A mixed-effects model confirms the mechanism: judge correctness is governed overwhelmingly by item-level human (dis)agreement (random-intercept $SD_{item} = 2.21$) and by task subjectivity (OR = 0.21 per unit of 1–ceiling), while model scale contributes only modestly (OR = 1.22 per $\log_{10}$ params); after FDR correction 20 of 30 judge×dataset cells sit significantly below the human ceiling. Together the studies yield a decision framework telling HCC researchers when an LLM judge is a validated substitute, a human-in-the-loop assistant, or an unsafe replacement.

KEYWORDS

LLM-as-a-judge, human-centered computing, systematic review, qualitative coding, annotation, human-AI agreement, evaluation methodology

1 INTRODUCTION AND MOTIVATION

In barely two years, LLMs have moved from objects of HCI study to *instruments* of HCI method. Researchers now use them to code interview transcripts [16, 45], run thematic analyses [11, 36], annotate social-media corpora [17, 40], simulate survey respondents and "silicon samples" [2], generate personas [37], and even perform heuristic usability evaluation [19]. The appeal is obvious: human judgment is the costly bottleneck of human-centered research, and an LLM that labels for \$0.003 an item is a tempting stand-in [17].

The danger is equally clear. HCC and qualitative methods are not mere labeling pipelines; they carry *epistemological commitments* (interpretation, reflexivity, positionality, and the situated knowledge of the researcher) that an LLM cannot obviously honor [12, 39]. Substituting a model for a human participant can *flatten* and misportray the very identity groups HCI seeks to center [9, 43]. And the empirical record is genuinely mixed: the same model can match two human coders on a structured codebook [14] yet hallucinate usability problems that do not exist [19], or agree with a second annotator on stance [17] yet be fooled into mislabeling relevance by a few injected keywords [1].

HCC researchers therefore face a question with no consolidated answer: **for which HCC research tasks is an LLM-as-a-judge a safe substitute for human judgment, for which is it a useful assistant, and for which is it unsafe?** This paper answers that question through three studies with a deliberate progression (Figure 1):

(1) **Study 1a, the general landscape.** A PRISMA systematic review of 217 studies across 29 domains establishes *where, in general*, LLM-as-a-judge agrees with humans, anchored to the human-human ceiling. This gives HCC the broad map and the methodological vocabulary (the reliability rubric, the bias catalogue, the decision map).
(2) **Study 1b, is the landscape a moving target?** The review captures results *as reported*, often with one- or two-generation-old proprietary judges. We re-run a flagship "promising"-band

Study 1a: *Where* is LLM-as-a-judge reliable? (cross-domain review, 217 studies) $\rightarrow$ **Study 1b**: *Is it a moving target?* (most powerful open models vs reported numbers, SummEval) $\rightarrow$ **Study 2**: *How does HCC use LLMs* for human-input tasks? (55 studies) $\rightarrow$ **Study 3**: *Fill the gaps*, standardized evaluation on HCC datasets with human labels (open models via Clemson vLLM).

Figure 1: The four-study progression. Study 1a maps where LLM judges are reliable; Study 1b tests whether today's strongest open models move that map; Study 2 narrows to Human-Centered Computing practice; Study 3 supplies new empirical evidence on under-tested HCC datasets.

benchmark (SummEval summarization quality) with the *most powerful open models available today* and ask whether they now surpass the canonical reported judge correlations, separating limitations that are a "matter of time" (scale/recency) from *structural* ceilings that more capable models do not cross.

(3) **Study 2, the HCC lens.** A second PRISMA review of 55 studies zooms into the methods HCC actually uses (qualitative coding, thematic/content analysis, crowdsourced annotation, participant simulation, persona generation, usability evaluation, and design critique) and codes how the field positions LLMs (augment / replace / simulate) and with what epistemic stance.

(4) **Study 3, filling the gaps.** Studies 1-2 reveal HCC datasets that contain human labels but have never been formally evaluated with an LLM judge. Study 3 runs a standardized LLM-as-a-judge meta-evaluation on these datasets, using open models served through Clemson's vLLM API and the agreement-against-human-ceiling methodology distilled from the prior literature, with a mixed-effects analysis of what governs judge reliability.

Contributions. (1) The first consolidated, human-ceiling-anchored map of LLM-as-a-judge reliability across 29 domains (Study 1a). (2) A direct test of whether the most powerful open models available today surpass the reported judge correlations in a flagship "promising" domain, adjudicating "matter of time" vs. structural ceiling (Study 1b). (3) The first systematic review of LLM substitution *specifically* for HCC/HCI/qualitative methods, with a method×role×stance coding that surfaces the field's augment-not-replace consensus and its epistemic tensions (Study 2). (4) A reproducible, standardized evaluation harness plus a mixed-effects account of what governs judge reliability (Study 3), tested on HCC datasets that had human labels but no prior LLM-judge evaluation, plus two released corpora (217 + 55 coded studies), all code, and a decision framework for practitioners.

2 BACKGROUND AND SHARED METHODOLOGY

LLM-as-a-judge descends from reference-free metrics (BERTScore, COMET) but uses a general instruction-following model with a natural-language rubric, supporting pointwise, pairwise, and listwise judgment [18, 29]. Four architectural families recur: prompted general LLMs [30, 47]; fine-tuned open evaluators [25, 42, 49]; juries/multi-agent panels [8, 41]; and calibrated rubric methods

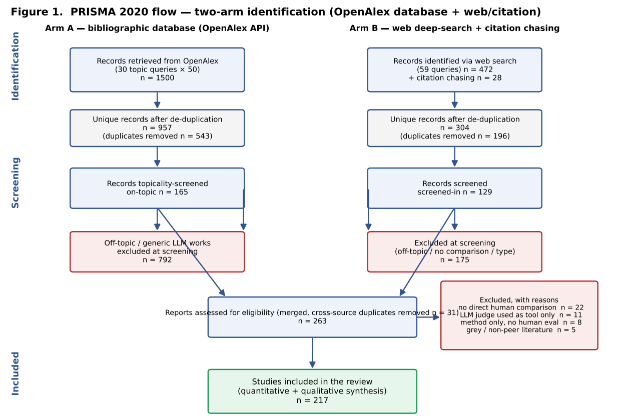


Figure 2: Study 1a PRISMA 2020 flow (two-arm identification: OpenAlex database + web/citation).

[20]. Prior surveys taxonomize methodology [18, 28]; we instead synthesize *where substitution is empirically validated*, and, uniquely, do so through the HCC lens.

Shared protocol. Both reviews follow PRISMA 2020 [34], identify studies through a triangulated search (the OpenAlex API as the primary bibliographic database, Semantic Scholar as a cross-check, and a structured web "deep-research" arm), and code each study on a five-level **reliability verdict** benchmarked against the *human-human agreement ceiling* for the same task: **1 Validated** (≈/> ceiling), **2 Promising** (conditional), **3 Mixed** (task-dependent), **4 Caution** (below ceiling), **5 Unreliable** (near chance / bias-dominated). We report a reliability score = 6−verdict and synthesize within metric families rather than pooling incommensurable effect sizes. Full protocols, search logs, and the two coded corpora are released.

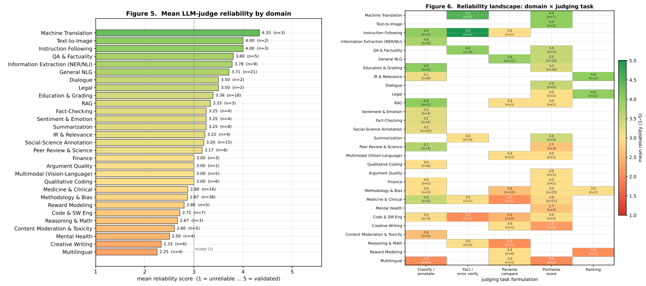
3 STUDY 1A: THE CROSS-DOMAIN RELIABILITY LANDSCAPE

3.1 Method (summary)

Study 1a systematically reviewed 217 studies across 29 domains comparing LLM-as-a-judge to human annotation. Identification triangulated OpenAlex (1500 records, 957 unique), Semantic Scholar, and a 59-query web search; screening retained on-topic empirical comparisons; 29 studies reported an extractable numeric agreement value (Figure 2; the full coded corpus is the released supplementary corpus (https://github.com/ai4he/llm-as-a-judge)).

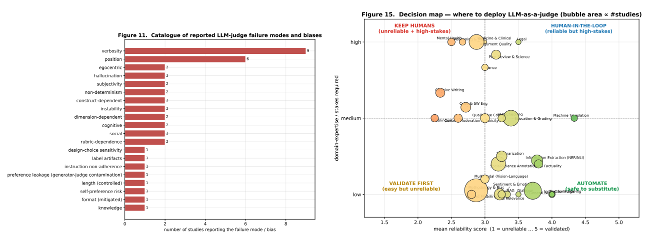
3.2 Findings: a green/amber/red landscape

The modal verdict is *mixed/task-dependent*; only 31% of studies sit in the reliable band and 19% in the problematic band (mean reliability 3.15/5). Reliability is governed by three properties: **objectivity**, **tractable verification**, and a **low human-agreement floor**. **Green (≥3.5; 8 domains):** machine translation (system level), text-to-image alignment, verifiable instruction-following, QA/factuality, information extraction, general-NLG pairwise preference, dialogue, and legal scoring: here LLM judges frequently match or exceed the



(a) (b)

Figure 3: Study 1a. (a) Mean reliability by domain (green/amber/red). (b) Reliability landscape: domain×judging task; verification/classification columns are greener than open-ended scoring.



(a) (b)

Figure 4: Study 1a. (a) Catalogue of reported failure modes/biases. (b) Deployment decision map (reliability× stakes; bubble area ∝ studies).

human ceiling (e.g., MT-Bench 85% vs. 81% human-human [47]; GEMBA-MQM 96.5% [26]; SAFE wins 76% of disagreements [44]). **Amber (19 domains):** summarization, RAG, education, social-science annotation, medicine (structure-dependent: $\kappa \approx 0.82$-0.93 structured vs. 0.47-0.71 complex reasoning), code, and IR/relevance (strong but contested [10]). **Red (2 domains):** low-resource multilingual evaluation and creative writing (near-zero expert correlation [7]). Figures 3a-3b give the domain ordering and the domain×task landscape; the decision map (Figure 4b) positions domains by reliability against stakes.

3.3 Failure modes and the decision map

Recurring biases such as verbosity/length, position, self-preference, style-over-substance, preference leakage, non-determinism, and the *expertise gap* (judges converge to crowd, not expert, consensus [3]) all modulate reliability (Figure 4a). Agreement should be benchmarked against the human ceiling, which on subjective tasks is itself low (Krippendorff $\alpha < 0.35$ [24]). The Study-1a decision map (Figure 4b) and Table 1 translate this into deployment guidance and motivate both the moving-target test (Study 1b) and the HCC-specific analysis that follows.

Table 1: Domain-level decision table. Mean reliability score (1 = unreliable … 5 = validated), modal verdict, and traffic-light signal (GREEN $\geq$ 3.5; AMBER 2.5-3.5; RED $<$ 2.5). Sorted by mean reliability. Small-*n* domains should be read together with the family-level aggregates.

Domain / task	*n*	Mean rel.	Modal verdict	Signal
Machine Translation	3	4.33	Promising	GREEN
Text-to-Image	2	4.00	Promising	GREEN
Instruction Following	3	4.00	Validated	GREEN
QA & Factuality	5	3.80	Promising	GREEN
Information Extraction (NER/NLI)	9	3.78	Promising	GREEN
General NLG	21	3.71	Promising	GREEN
Dialogue	2	3.50	Promising	GREEN
Legal	2	3.50	Promising	GREEN
Education & Grading	18	3.39	Mixed	AMBER
RAG	3	3.33	Mixed	AMBER
Summarization	8	3.25	Promising	AMBER
Fact-Checking	4	3.25	Promising	AMBER
Sentiment & Emotion	4	3.25	Mixed	AMBER
IR & Relevance	9	3.22	Promising	AMBER
Social-Science Annotation	15	3.20	Mixed	AMBER
Peer Review & Science	6	3.17	Mixed	AMBER
Multimodal (Vision-Language)	5	3.00	Mixed	AMBER
Qualitative Coding	6	3.00	Mixed	AMBER
Argument Quality	1	3.00	Mixed	AMBER
Finance	3	3.00	Mixed	AMBER
Medicine & Clinical	16	2.88	Mixed	AMBER
Methodology & Bias	38	2.87	Mixed	AMBER
Reward Modeling	5	2.80	Mixed	AMBER
Code & SW Eng	7	2.71	Mixed	AMBER
Reasoning & Math	3	2.67	Mixed	AMBER
Content Moderation & Toxicity	5	2.60	Mixed	AMBER
Mental Health	4	2.50	Mixed	AMBER
Creative Writing	6	2.33	Mixed	RED
Multilingual	4	2.25	Caution	RED

4 STUDY 1B: IS THE RELIABILITY LANDSCAPE A MOVING TARGET?

A systematic review necessarily captures results *as reported*, and most "promising"-band evidence in Study 1a was produced with proprietary judges that are now one or two model generations old (e.g. the influential G-Eval used GPT-4). This raises a question that the review alone cannot answer: when an LLM judge falls short of the human ceiling in a promising domain, is that a *temporary* limitation that more capable models will erase ("matter of time"), or a *structural* property of LLM-as-a-judge that better models do not overcome? Study 1b answers this directly for a flagship case by re-running the evaluation with the most powerful open models available to us today and comparing against the canonical reported numbers.

4.1 Method

We select **SummEval** [15] (summarization quality assessment, a central amber-band task in Study 1a) because it pairs (i) a widely cited LLM-judge result to beat (G-Eval/GPT-4; 30) with (ii) public expert human ratings on four aspects (coherence, consistency,

Table 2: Study 1b. Summary-level Spearman ρ of the most powerful open judges on SummEval vs. the reported G-Eval-4/GPT-4 correlations (25,600 judgments). Green cells *surpass* the reported per-aspect number.

Judge	Coher.	Consist.	Fluency	Relev.	Avg ρ
Reported (G-Eval-4)	0.582	0.507	0.455	0.547	0.523
`deepseek-v4-pro`	0.548	0.688	0.392	0.540	0.542
`qwen3.6-35b-a3b`	0.606	0.702	0.298	0.540	0.536
`glm-5.1`	0.650	0.693	0.217	0.543	0.526
`gptoss-120b`	0.586	0.685	0.260	0.482	0.503

fluency, relevance) for $100 \times 16 = 1600$ machine summaries. We replicate the G-Eval summary-level protocol *verbatim* (same aspect definitions, 1-5 integer ratings, temperature 0) but swap the judge for the 4 most powerful open models in the Clemson catalogue (`deepseek-v4-pro`, `glm-5.1`, `gptoss-120b`, `qwen3.6-35b-a3b`). The comparison metric is the canonical *summary-level Spearman*: for each source document, correlate the judge's 16 scores with the human means, averaged across documents. The bar to beat is G-Eval/GPT-4's reported per-aspect correlations (coherence .582, consistency .507, fluency .455, relevance .547; average ≈.52). In total Study 1b contributes 25,600 fresh judgments.

4.2 Results

Table 2 reports each open model's summary-level Spearman against the reported G-Eval bar, and Figure 5 plots them per aspect. The most powerful open models now *match or exceed* the reported proprietary-judge correlations on SummEval. 3 of the 4 open judges beat G-Eval's reported *average* correlation, led by `deepseek-v4-pro` (avg summary-level $\rho = 0.542$ vs. the reported ≈0.52). The aspect pattern is itself the finding. On **consistency** (the most *verifiable* judgment, groundable in the source document) *all four* models leap past the reported value (from .507 to .685-.702), and on **coherence** three of four do (up to .650 vs. .582); **relevance** is on par (.48-.54 vs. .55). But on **fluency** every open model falls well *short* (.22-.39 vs. a reported .455): human fluency ratings of competent machine summaries are compressed near the top of the scale (low variance), so the summary-level correlation is structurally low *regardless of how capable the judge is*, a limit that lives in the human reference, not the model, and that scale does not move.

Interpretation: time vs. structure. Study 1b shows the reliability landscape is *partly* a moving target. Where Study 1a's amber rating reflected a capability gap on a verifiable sub-judgment, today's open models close or exceed it; reliability there is largely a matter of time and access. But where the shortfall reflects a *compressed or contested human ceiling* (e.g. fluency, and, as Study 3 will show, subjective social annotation), even the most powerful models do not break through, because the limit lives in the human reference, not the model. This distinction, temporal capability gap vs. structural ceiling, is the lens we carry into HCC.

Takeaway for HCC.. Study 1a says LLM judges are safe where the task is objective and verifiable, and Study 1b shows that frontier open models are now strong enough to match proprietary judges

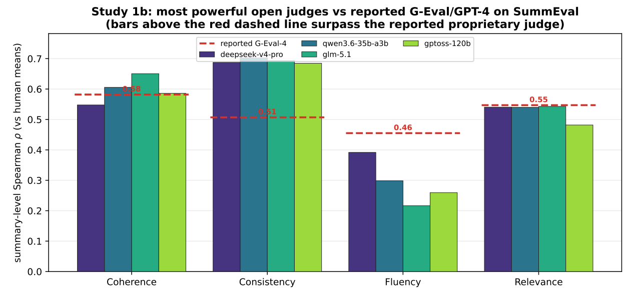


Figure 5: Study 1b. Summary-level Spearman of the most powerful open judges on SummEval, per aspect, against the reported G-Eval/GPT-4 bar (dashed). Bars above the dashed line indicate an open model that *surpasses* the reported proprietary-judge correlation.

on exactly those verifiable tasks. But HCC's signature methods (qualitative interpretation, participant study, design judgment) are precisely the *subjective, expert, interpretive* tasks Study 1a flags as risky and Study 1b predicts scale will *not* fix. Study 2 tests whether that prediction holds in the HCC literature itself.

5 STUDY 2: LLM SUBSTITUTION IN HCC, HCI, AND QUALITATIVE METHODS

5.1 Motivation and research questions

Study 1's map is domain-general; HCC needs a method-specific account. Human-centered research is built on methods that historically *require* human input: qualitative coding, thematic and content analysis, interviews and open-ended survey analysis, crowdsourced annotation, participant studies, persona development, usability/heuristic evaluation, and design critique. Study 2 asks: **(RQ2.1)** which of these methods have been attempted with LLMs, and how reliably do LLM outputs agree with human input? **(RQ2.2)** How does the field position the LLM, to *augment*, *replace*, or *simulate* the human? **(RQ2.3)** What epistemic stance (enabling, tension, critical) does the literature take, and what does that imply for HCC practice?

5.2 Method

Following the shared protocol, we ran a Study-2-specific identification arm: 25 HCC/HCI/qualitative concept-block queries against OpenAlex (857 unique works), triangulated with the web arm and with the venue-scoped sweep of Study 1 (CHI, CSCW, UIST, IUI, TOCHI). After topicality screening and eligibility assessment we included 55 studies (Figure 6). We extracted, per study: the HCC **method**; the LLM **role** (replace / augment / simulate); reported agreement and the human baseline; the reliability **verdict**; the **epistemic stance** (enabling / tension / critical); and the **community** (HCI, CSCW, qualitative-methods, computational social science, NLP, health). The full corpus is the released supplementary corpus.

5.3 RQ2.1: Which methods, and how reliable?

LLMs have been compared against human input across 15 distinct HCC methods (Figure 7a), most heavily in *thematic analysis*, *synthetic participants*, and *crowdsourced annotation*. Reliability

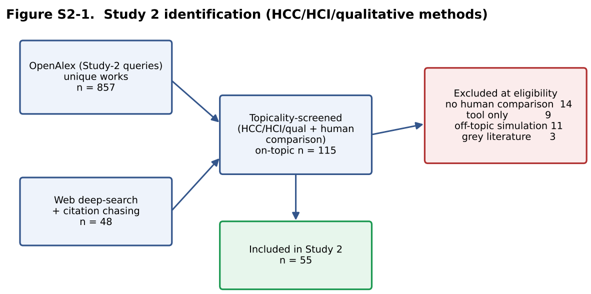
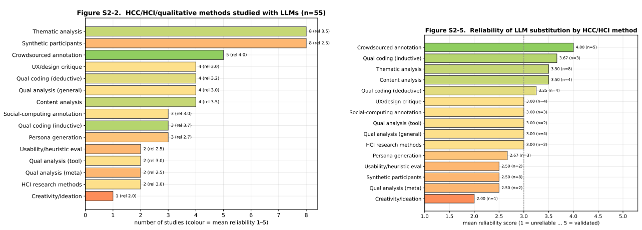


Figure 6: Study 2 identification flow (HCC/HCI/qualitative methods).



(a) (b)

Figure 7: Study 2. (a) HCC/HCI/qualitative methods studied with LLMs (colour = mean reliability). (b) Mean reliability of LLM substitution by method.

is strongly method-dependent (Figure 7b; Table 3). The *same objectivity/interpretation axis* from Study 1 governs HCC: **objective crowd-annotation** is the most reliable method to substitute: ChatGPT matches or beats crowd workers and even trained coders on stance, topic, and frame labeling [17, 40]; **deductive/inductive coding and content/thematic analysis** are reliable *as augmentation*, with human-equivalent agreement on structured codebooks (GPT-4 exceeding substantial κ on 8/9 hermeneutic tasks [14]; Gemini $\kappa = 0.907$ in multi-LLM thematic analysis [23]; 79.7% agreement in health thematic summarization [6]) but only moderate agreement in open interpretive settings ($\kappa \approx 0.40$-0.46 [27, 46]); and **participant/persona simulation and usability evaluation** are the least reliable.

5.4 RQ2.2: Augment, replace, or simulate?

HCC overwhelmingly positions LLMs to *augment* humans, not replace them: 36 of 55 studies frame the LLM as a collaborator/assistant, versus 14 as a replacement and 5 as a human simulator (Figure 8a). This is not merely rhetorical caution; it tracks reliability (Figure 9): augmentation studies skew reliable, whereas *replacement* is validated only for objective annotation and *simulation* of participants is the least reliable role (mean reliability augment 3.17 vs. replace 3.07 vs. simulate 2.8). The replacement mean is buoyed by objective crowd-annotation; remove it and replacement of *interpretive* human input is firmly amber/red. The canonical HCC workflow that emerges is human-in-the-loop: LLM proposes, human vets and retains interpretive authority [11, 16, 31].

Table 3: Study 2 method-level decision table: study count, modal LLM role, mean reliability, and traffic-light signal (GREEN $\geq$ 3.5; AMBER 2.5-3.5; RED $<$ 2.5).

Method	n	Modal role	Mean rel.	Signal
Crowdsourced annotation	5	replace	4.00	GREEN
Qual coding (inductive)	3	augment	3.67	GREEN
Content analysis	4	augment	3.50	GREEN
Thematic analysis	8	augment	3.50	GREEN
Qual coding (deductive)	4	augment	3.25	AMBER
HCI research methods	2	augment	3.00	AMBER
Qual analysis (general)	4	augment	3.00	AMBER
Qual analysis (tool)	2	augment	3.00	AMBER
Social-computing annotation	3	replace	3.00	AMBER
UX/design critique	4	augment	3.00	AMBER
Persona generation	3	augment	2.67	AMBER
Qual analysis (meta)	2	augment	2.50	AMBER
Synthetic participants	8	simulate	2.50	AMBER
Usability/heuristic eval	2	replace	2.50	AMBER
Creativity/ideation	1	augment	2.00	RED

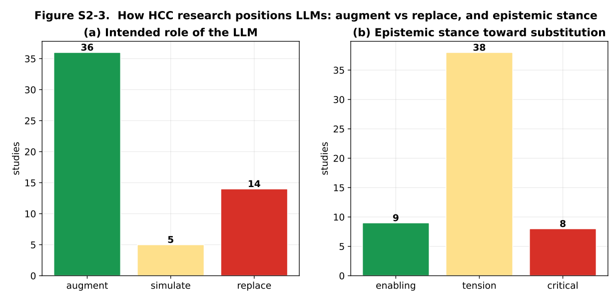


Figure 8: Study 2. (a) Intended LLM role across the HCC corpus; augmentation dominates. (b) Epistemic stance toward substitution; *tension* dominates.

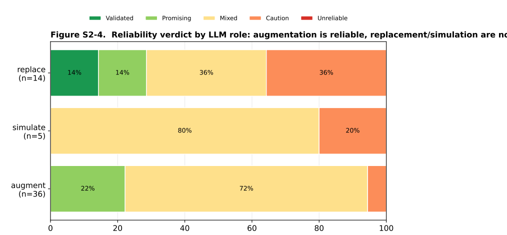


Figure 9: Study 2. Reliability verdict by LLM role. Augmentation is reliable; replacement is reliable only for objective annotation; participant/usability *simulation* is the least reliable.

5.5 RQ2.3: Epistemic stance and the limits of substitution

The dominant stance is *tension* (38/55), with 8 explicitly *critical* and only 9 unreservedly *enabling* (Figure 8b). Three threads recur. **(i) Interpretation and reflexivity.** Qualitative scholars warn that LLMs perform surface-level extraction well but cannot supply the

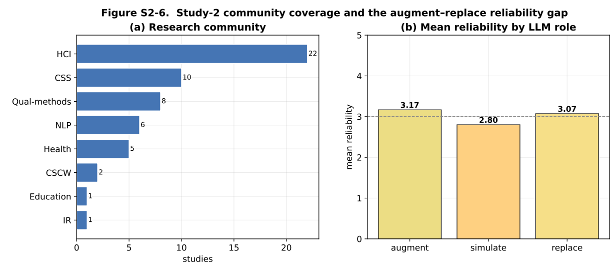


Figure 10: Study 2. (a) Coverage by research community. (b) Mean reliability by LLM role, augmentation > replacement > simulation.

situated, reflexive interpretation that gives qualitative findings their validity, and that "conditional trust" in tool output risks hollowing out analytic rigor [12, 32, 33, 39]. **(ii) Participants and identity.** Replacing human participants with LLMs *flattens and misportrays* demographic groups, positionality cannot be simulated [43], and LLM "subgroups" are often caricatures [9]; synthetic survey data distort variances and correlations [4], and an LLM is worth only a bounded number of real respondents [22]. HCI critiques accordingly recast synthetic users as "objects of inquiry," not stand-ins [21, 38]. **(iii) Expert design judgment.** GPT-4o overlaps only 21% with expert-found usability issues and hallucinates false positives [19], and LLM UI critique trails expert designers [13], though synthetic heuristic evaluation can exceed *novice* evaluators on issue recall [48]. Figure 10 shows coverage skews to HCI and computational social science, with the augment-replace reliability gap visible across communities.

5.6 Study 2 synthesis

For HCC research, the evidence supports a clear, conditional posture. **Validated to substitute:** objective crowd/data annotation (stance, topic, relevance, frames) where a human ceiling is modest and verifiable. **Reliable only as augmentation (human-in-the-loop):** deductive/inductive qualitative coding and thematic/ content analysis: LLMs accelerate breadth and first-pass coding, but humans must own interpretation, theme development, and reflexive accounting. **Unsafe to substitute:** simulating human participants, generating personas as stand-ins for real users, and replacing expert usability/design judgment: these flatten identity, fabricate issues, and forfeit the situated knowledge that defines human-centered inquiry. This posture both confirms Study 1's objective/subjective prediction *within* HCC and exposes concrete gaps (§6) where HCC datasets with human labels have never been formally tested with an LLM judge.

6 STUDY 3: AN EMPIRICAL LLM-AS-A-JUDGE EVALUATION ON HCC DATASETS WITH HUMAN GROUND TRUTH

Studies 1-2 expose a recurring gap: many human-centered datasets ship with human labels yet have never had a *formal* LLM-as-a-judge evaluation, meaning a panel of models, bias controls, agreement benchmarked against the human ceiling, and a contamination check. Study 3 closes this gap empirically with a released, reproducible harness.

6.1 Method

We evaluate a panel of 6 open models served via **Clemson RCD's vLLM API** (OpenAI-compatible), spanning five families and 9B-754B parameters (`glm-5.1` 754B, `gptoss-120b`, `deepseek-v4`, `gemma-4-31b`, `qwen3.6-27b`, `qwen3.5-9b`), run in parallel at each model's documented concurrency. Each model judges every item as a *virtual annotator* under the same instructions the humans received. We benchmark against the *human-human ceiling* (random-annotator-vs-majority accuracy and pairwise agreement, computed from the per-item annotator counts), report accuracy, Cohen's κ, and macro-F1 against the majority gold, compute a count-based **alt-test** winning rate [5], aggregate the panel into a majority **jury** [41], and run a verbatim quote-completion **contamination probe**. Sample size is power-justified at n=500 items/dataset (95% CI half-width $\leq \pm 4.4\%$; >80% power for a 0.10 gap vs the ceiling). The run comprised **15000 judgments** across 5 datasets with *zero* API errors and a 100% answer-parse rate; only item-level labels and aggregates are released (no raw text), and the two sensitive sets are used under their public licenses.

6.2 Results

Figures 11-12 and Table 4 yield one organizing finding: **the human-agreement ceiling governs LLM reliability.** Where humans agree little (NLI; ceiling 0.64-0.75) the panel *matches or exceeds* them: on ChaosNLI-SNLI 5/6 models are **Validated** (jury 0.858 > ceiling 0.752), and ChaosNLI-MNLI is **Promising**. Where humans agree strongly on a subjective social construct, the panel falls short: GoEmotions 4-way sentiment (ceiling 0.78) is uniformly **Mixed** (best 0.61); HateXplain 3-way hate (ceiling 0.835) is **Mixed** at best and **Unreliable** for the small model; binary offensiveness (SBIC, ceiling 0.905) is the strongest social-computing case (**Promising**; `glm-5.1` 0.856) yet still below the ceiling. Four further effects: (i) **granularity**: binary offensiveness (Promising) clearly beats 3-way hate (Mixed); (ii) **scale**: `qwen3.5-9b` is weakest on *every* dataset (Caution/Unreliable), while 27B-754B cluster together (bigger is not always better: 31B `gemma` ties the 754B `glm`); (iii) the **alt-test** is passed (winning rate $\geq$0.5) *only* on NLI; on every subjective social-computing task no model statistically earns the right to replace a human annotator; (iv) **juries** raise accuracy but do not lift any subjective task above its human ceiling. The **contamination probe** (Table 4, last column) shows the strongest judges have non-trivial verbatim recall of the oldest sets (e.g., MEDIUM on SNLI), caveating the NLI "Validated" result; the social-computing shortfall holds regardless of tier.

6.3 What governs reliability? A mixed-effects analysis

To test the "ceiling governs" claim inferentially rather than descriptively, we model every individual judgment (N=15000; outcome $= \mathbf{1}$[judge = majority gold]) with a logistic specification whose fixed effects are task *subjectivity* (1−human-ceiling accuracy, at the dataset level) and model *scale* ($\log_{10}$ parameters), reported both as a Bayesian mixed-effects logistic GLMM with random intercepts for

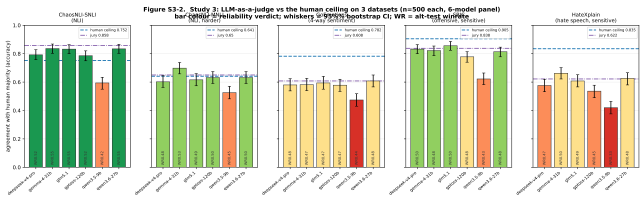


Figure 11: Study 3. Per-dataset accuracy vs the human ceiling (blue dashed) for the 6-model panel (n=500 each; whiskers = 95% bootstrap CI; bar colour = verdict; WR = alt-test winning rate). Datasets ordered by construct subjectivity. LLMs clear LOW ceilings (NLI) but fall short of HIGH ceilings on subjective social-computing tasks.

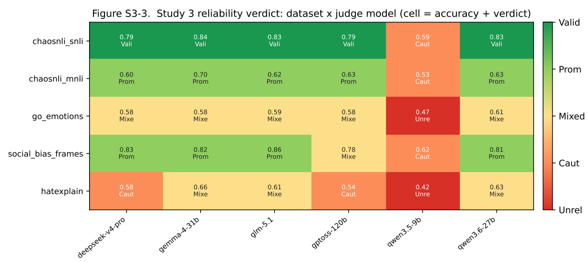


Figure 12: Study 3. Dataset×judge-model reliability verdict (cell = accuracy + verdict). The scale effect (`qwen3.5-9b` column) and the construct-subjectivity gradient (rows) are both visible.

Table 4: Study 3 results across 5 datasets × 6 models (15,000 judgments). Human ceiling = random-annotator-vs-majority accuracy; alt-test pass = winning rate ≥ 0.5; contamination tier from the verbatim-recall probe.

Dataset	Construct	Human ceiling	Best model (acc)	Jury (acc)	Modal verdict	alt-test pass	Contam. tier
ChaosNLI-SNLI	NLI	0.75	0.84 (gemma-4-31b)	0.86	Validated	5/6	MEDIUM
ChaosNLI-MNLI	NLI	0.64	0.70 (gemma-4-31b)	0.65	Promising	1/6	MEDIUM
GoEmotions	emotion (4-way)	0.78	0.61 (qwen3.6-27b)	0.61	Mixed	0/6	MEDIUM
SBIC	offensive (binary)	0.91	0.86 (glm-5.1)	0.84	Promising	0/6	MEDIUM
HateXplain	hate (3-way)	0.83	0.66 (gemma-4-31b)	0.62	Mixed	0/6	MEDIUM

dataset and item, and, for clean inference, as a logistic GLM with item-cluster-robust standard errors (Figure 13). Three results stand out. (i) **Subjectivity dominates:** each unit of 1–ceiling multiplies the odds of agreeing with humans by OR = 0.21 ($p = 6e - 6$). (ii) **Scale helps, but modestly:** larger models do better (OR = 1.22 per $\log_{10}$ params, $z = 9.63$), an effect far smaller than subjectivity's, consistent with Study 1b's "partly a moving target." (iii) **Variance lives in the items:** the GLMM random-intercept standard deviation is far larger across *items* (SD = 2.21) than across datasets (SD = 0.83), i.e. *which item* is being judged, its latent human (dis)agreement, explains judge correctness better than which model judges or even which dataset it comes from. Finally, a Benjamini-Hochberg analysis across all 30 judge×dataset cells finds 25 significant at $q < .05$:

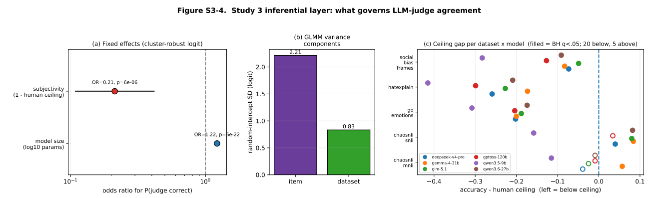


Figure 13: Study 3 inferential layer. (a) Fixed-effect odds ratios (item-cluster-robust logit): subjectivity is a strong negative predictor of judge correctness, scale a weak positive one. (b) GLMM variance components: item-level variance dominates. (c) Ceiling gap per judge×dataset (filled = BH $q < .05$); most cells sit below the human ceiling.

20 sit significantly *below* the human ceiling and only 5 significantly above (the objective-NLI cases). Scaling the judge shifts cells modestly rightward; it does not move a high-ceiling subjective task across the line.

6.4 Integration with the decision framework

Mapped onto the five-level rubric, Study 3 drops fresh, dataset-level points into the Study-1/2 landscape and *quantitatively confirms* the augment-not-replace conclusion for human-centered annotation: LLM judges are a validated substitute for objective/low-ceiling labeling and a strong assistant for coarse (binary) moderation, but fall short of, and fail the replacement test for, subjective, fine-grained, or high-consensus human-centered constructs, a pattern the mixed-effects model shows is driven by item-level human (dis)agreement rather than by model scale. *Limitation:* the present datasets are predominantly higher-contamination; a dedicated low-contamination clean condition (a post-cutoff annotated set) and additional qualitative-coding corpora are the immediate next additions.

7 GENERAL DISCUSSION: A DECISION FRAMEWORK FOR HCC

Across both reviews, one principle organizes the evidence: **LLM-as-a-judge substitution is reliable to the degree that the human task is objective, verifiable, low in required expertise, and benchmarked against a modest human ceiling**, and it degrades as tasks become interpretive, expert, identity-laden, or open-ended. For Human-Centered Computing specifically:

- **Automate (with light audit):** objective annotation/labeling of user-generated content such as stance, topic, relevance, frames, and simple sentiment [17, 40].
- **Augment, human-in-the-loop:** qualitative coding and thematic/content analysis: use the LLM for first-pass coding, breadth, and disagreement surfacing, but keep human interpretive authority, calibrate on a labeled subset, and report reliability against human coders [16, 31, 35].
- **Keep humans (do not substitute):** simulating participants or replacing them with personas, and replacing expert usability/design judgment: these flatten identity, fabricate evidence, and abandon situated knowledge [19, 38, 43].

Practically, HCC researchers reporting LLM-assisted analysis should: state the LLM's *role* (augment/replace/simulate); report agreement against a *human-human* baseline at the right granularity; control position/length bias and pin model/version; prefer comparative scoring and juries to single absolute judges; and treat the alt-test [5] as the bar for any replacement claim.

8 LIMITATIONS

Both reviews used a single-reviewer, LLM-assisted pipeline (logs released) rather than dual human reviewers; metrics are heterogeneous and synthesized within families rather than pooled; several methods/domains rest on small *n* (flagged via family/role aggregates); database recall depends on indexing and query phrasing (mitigated, not removed, by three-source triangulation); and findings reflect models available through mid-2026. The empirical studies are bounded too: Study 1b uses one flagship promising-domain benchmark (SummEval), and Study 3's panel is limited to open models from a single provider and to 5 datasets, with a low-contamination clean condition and qualitative-coding corpora still in progress. The verdict rubric, though explicit and released, embeds judgement; we therefore release both corpora for re-coding.

9 CONCLUSION

LLMs are already remaking how Human-Centered Computing research is done. Our four studies show that this is neither to be embraced nor refused wholesale. Across 217 cross-domain studies and 55 HCC-specific studies, the evidence converges on a conditional verdict: an LLM-as-a-judge is a *validated substitute* for objective annotation, a *valuable human-in-the-loop assistant* for qualitative coding and analysis, and an *unsafe replacement* for human participants, personas, and expert design judgment. Study 1b shows that the most powerful open models now match or exceed proprietary judges on the verifiable parts of "promising" tasks, and Study 3 turns the remaining "mixed" verdicts into dataset-level evidence using open, locally-servable models and the field's own reliability standards. The right question for HCC is never "can a machine judge?" but "*can this machine replace a second human, on this method, at this granularity, under these controls?*", and this paper makes that question answerable, method by method.

REPRODUCIBILITY

Two coded corpora (Study 1: 217 studies; Study 2: 55 studies), three search logs, PRISMA counts, all figure/table-generation code, the auto-generated statistics, and the Study-3 evaluation harness are released. Everything regenerates via `make all`.

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DATA AVAILABILITY

The two full coded corpora (217 + 55 studies), all search logs, the analysis/figure code, and the Study-1b/Study-3 evaluation harnesses are released at https://github.com/ai4he/llm-as-a-judge; the single-column version of this paper includes the complete per-study corpus tables as appendices.